

Onboarding Experiment Investigation

This assignment consists of a single data file capturing the results of an A/B test.

Files

`experiment_results.csv`: one row per user who went through signup last month. Columns:

column	meaning
<code>user_id</code>	unique identifier
<code>segment</code>	how the user found the product: <code>organic</code> , <code>paid_search</code> , <code>referral</code> , <code>app_store</code> , <code>influencer</code>
<code>variant</code>	which onboarding flow the user saw: <code>control</code> (old flow) or <code>treatment</code> (new flow)
<code>converted</code>	<code>1</code> if the user completed signup, <code>0</code> if they dropped off

The Problem

The growth team shipped a new onboarding flow to a random subset of new users last month. Looking at the overall numbers, leadership is ready to roll the new flow out to everyone. The topline conversion rate looks clearly better for `treatment` than `control`.

Before that rollout happens, you've been asked to sanity-check the claim. Is the new flow actually better, for everyone, some users, or is something else going on? Your job is to investigate the data and report back with numbers, not a gut feeling.

Questions

1. What is the overall (naive) difference in conversion rate between `treatment` and `control` across the whole file? Report the exact value in percentage points, and how many users are in each variant.
2. Break the same comparison down by `segment`. For each segment, report sample size and conversion rate for both variants. Is there a segment whose lift looks impressive but that you would NOT trust, and why not?
3. Compute a **mix-adjusted overall lift**: take each segment's own treatment-vs-control lift, weight it by that segment's share of the *total* user population (not by how many of that segment happen to be in `treatment`), and sum. Report the exact number to 2 decimal places (percentage points). In 2-3 sentences, explain why this number is different from your answer to Question 1.
4. Based on everything above, is there a segment where you believe the new onboarding flow has a real, meaningful positive effect? State which one (or "none") and the evidence that convinced you.
5. **Bonus:** Look at what fraction of each segment ended up in `treatment` vs `control`. Does anything about how users were assigned look off to you?

Submit

- `ANSWERS.md` containing your answers to all 5 questions, with the exact numbers and how you got them.
- `answers.json` with the following fields (used for automated fact-checking of your numeric answers; this doesn't replace `ANSWERS.md`, it just lets us verify the exact numbers quickly):

```
{  
  "q1_naive_lift_pp": 0.0,  
  ...  
}
```

```
"q1_n_control": 0,  
"q1_n_treatment": 0,  
"q2_untrustworthy_segment": "",  
"q3_mix_adjusted_lift_pp": 0.0,  
"q4_real_effect_segment": ""  
}
```

- Any scripts or code you used to analyze the data.
- A short section describing your investigation process (5–10 bullet points), including anything you checked that turned out to be a dead end.

Ground rules

- You may use any language or tools, including AI assistants. Your submission should reflect your own investigation and reasoning. You should be able to walk through your analysis and explain any of your numbers live.
- Do not submit `experiment_results.csv`.
- Reading the CSV requires nothing beyond a standard library (Python's `csv` module, or `pandas` if you have it installed; either is fine).